Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Laser-Structured High-Silicon Thick Anode Technology Enabling ≥300 Wh·kg^{-1} and 5C-Capable Lithium-Ion Drone Batteries			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical / Electronics		Electrical / Electronics	
	Medium Category	**Small Category**	**Medium Category**	**Small Category**
	Batteries	Secondary Batteries	Batteries	Battery Materials
Concept of Proposed Technology	The proposed technology is a laser-architected silicon-based anode designed to enable simultaneously high energy density and high-power capability in lithium-ion batteries, specifically targeting drone applications. The core concept is to combine high-silicon-content anode materials (Si/C or SiOx, 10–30 wt% Si) with three-dimensional laser patterning applied to thick electrodes (40–100 μm). While high silicon loading and increased electrode thickness are effective in raising gravimetric energy density, they inherently degrade rate capability and cycle life due to increased ionic transport resistance and silicon-induced mechanical stress. The proposed technology addresses these limitations through engineered electrode architectures rather than changes in cell chemistry. Laser patterning introduces controlled three-dimensional features—such as channels and segmented domains—within the anode, which:			

	- Reduce effective lithium-ion diffusion length and electrode tortuosity, - Improve electrolyte wetting throughout thick electrodes, - Alleviate in-plane mechanical stress caused by silicon volume expansion, and - Preserve electronic conductivity pathways under high-rate operation. As a result, the technology enables areal capacities of 4–6 $mAh \cdot cm^{-2}$, cell-level energy densities ≥300 $Wh \cdot kg^{-1}$, and stable discharge at rates up to 5C, while maintaining ≥80% capacity retention over 100 cycles in full-cell pouch configurations. Importantly, the technology is designed to be compatible with roll-to-roll electrode manufacturing, with laser structuring implemented as an inline, scalable process step. This makes the proposed solution suitable not only for laboratory validation but also for industrial deployment. Potential applications extend beyond drones to UAVs, robotics, defense systems, and other high-power portable energy storage devices where both energy density and power output are critical.
Rationale for Joint R&D Project	The proposed technology cannot be realized through material development or process optimization alone. It requires the tight integration of advanced AI-/ML-based electrode architecture design, laser process engineering **(University of Toronto, Canada)**, and industrial battery manufacturing **(Remplir Inc., Republic of Korea)**, which motivates the need for a joint R&D project. The University of Toronto brings unique expertise in laser-based electrode structuring, including simulation-driven process optimization, data- and ML-assisted architecture design, and integration of laser processing into roll-to-roll manufacturing environments. These capabilities are essential for designing reproducible, high-throughput, and performance-optimized electrode architectures. Remplir Inc. (Remplir) contributes industrial-grade silicon anode manufacturing capabilities, including slurry formulation, thick-electrode coating, calendaring, and full-cell pouch battery assembly, as well as direct access to commercial drone battery markets. This ensures that the developed technology is evaluated under realistic manufacturing constraints and validated at the cell level with clear commercialization pathways. Collaboration is therefore essential to: - Bridge the gap between academic structured-electrode concepts and industrial battery production, - Rapidly iterate between electrode design, laser processing, and full-cell performance feedback, and - De-risk the scale-up and commercialization of high-silicon anode technologies. Through this joint R&D framework, the project establishes a process-driven and architecture-centric approach to overcoming the energy–power trade-off in lithium-ion drone batteries, delivering both scientific advancement and tangible industrial impact.
Key Words	1. Silicon-Based Anodes 2. High C-rate Drone Batteries 3. Roll-to-Roll Laser Processing 4. 3D-Architectured Thick Electrode

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	The objective of this joint R&D project is to develop and validate **laser-structured silicon-based anodes** that enable simultaneously high energy density and high-power capability (high C-rate) in lithium-ion batteries for drone applications, while maintaining manufacturability and commercial relevance. Specifically, the project aims to: - Develop Si/C and SiOx-based anodes with elevated silicon content (10–30 wt%) suitable for thick-electrode configurations. - Introduce three-dimensional laser-patterned electrode architectures that mitigate silicon-induced mechanical degradation and enhance ionic/electronic transport in thick anodes. - Integrate laser structuring into roll-to-roll (R2R)-compatible electrode processing, supported by simulation, inline sensing, and data-driven optimization. - Demonstrate the performance advantages of structured silicon anodes in full-cell pouch batteries paired with commercial cathodes (LCO, NCA, NCM), under drone-relevant operating conditions. - Establish a scalable and industrially transferable process framework for high-silicon anode manufacturing.			
R&D Target Specifications	**(1) Anode-Level Targets** - Silicon content in anode active material: 10–30 wt% - Electrode thickness: 40–100 μm - Areal capacity: ≥4–6 mAh·cm^{-2} - Electrode thickness swelling after cycling: ≤3% **(2) Cell-Level Performance Targets (Full-cell with pouch type)** - Cell-level gravimetric energy density: ≥300 Wh·kg^{-1} - Discharge rate capability: up to 5C - Cycle life: ≥100 cycles with ≥80% capacity retention - Cathode compatibility: LCO, NCA, NCM **(3) Process and Manufacturing Targets** - Laser patterning precision (depth variation): ≤±5 μm - Compatibility with roll-to-roll electrode processing - Reproducible structured electrode fabrication suitable for scale-up			
R&D Plan	The R&D activities for this project will be executed through a phased, iterative approach, combining electrode manufacturing, laser structuring, and full-cell validation. **Phase 1: Baseline Electrode Development and Characterization (1st year)** **Milestone:** Validated baseline thick silicon anodes with quantified limitations - Fabrication of Si/C and SiOx anodes with 10–30 wt% silicon and thicknesses of 40–100 μm. - Optimization of slurry formulation, coating, drying, calendaring, and thermal processing. - Baseline electrochemical and mechanical characterization of unstructured thick electrodes. - Definition of initial performance gaps in rate capability and cycle life			

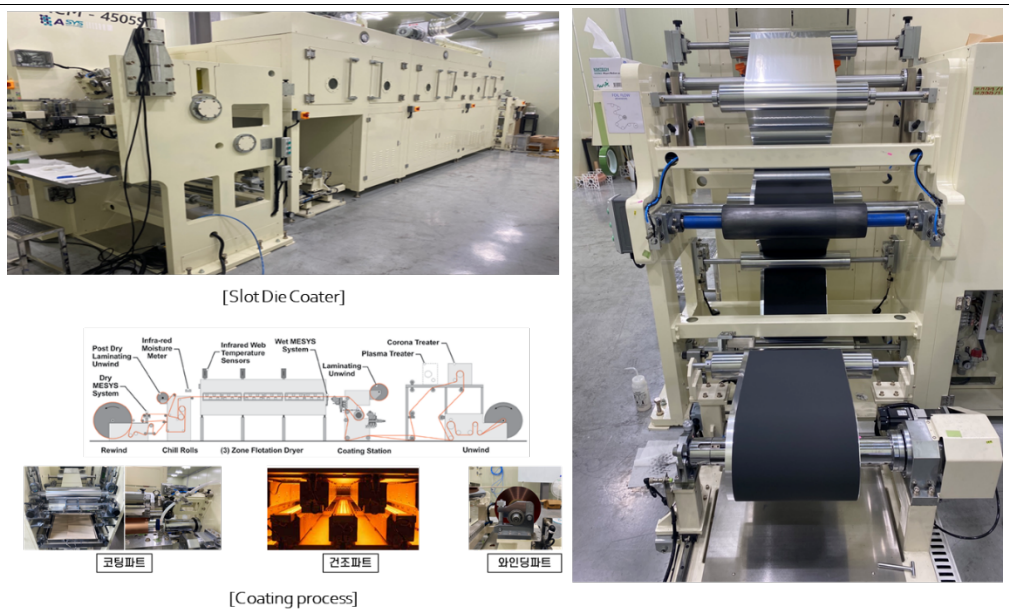

Figure 1. Industrial R2R-compatible electrode processing

Phase 2: Laser Architecture Design and Process Optimization (2nd year)
Milestone: Optimized laser-patterned anodes meeting transport and mechanical stability targets.

- Data- and ML-assisted design of electrode architectures linking geometry to performance metrics.
- Simulation-driven optimization of laser parameters (pattern geometry, spacing, depth, throughput).
- Laser structuring of electrodes using R2R-compatible processing concepts.
- Inline sensing and adaptive control to ensure patterning uniformity and reproducibility.

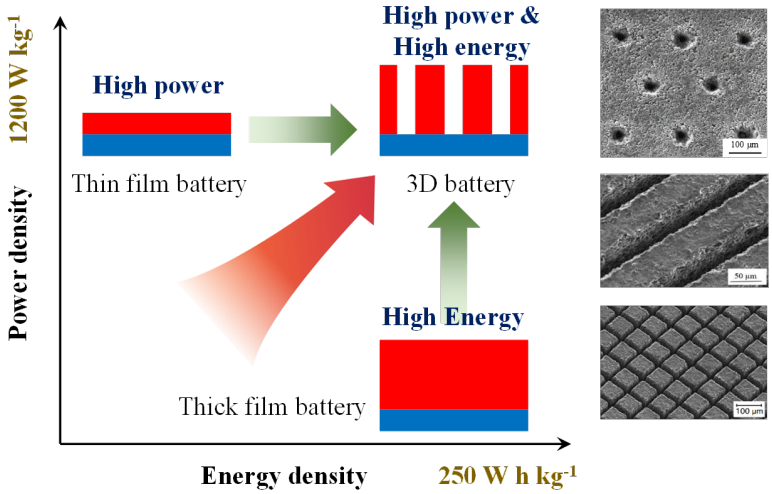

Figure 2. Schematic representation of the effect of electrode architecture development on energy and power density improvement.

Phase 2: Full-Cell Integration and Performance Validation (3rd year)
Milestone: Demonstration of ≥300 Wh·kg^{-1}, 5C-capable, 100-cycle-stable drone battery cells.

- Assembly of pouch-type full cells using laser-structured anodes and commercial cathodes (LCO, NCA, NCM).
- Evaluation of rate capability (up to 5C), cycle life (≥100 cycles), and dimensional stability.

- Iterative feedback between cell-level performance and electrode/process design.
- Assessment of scalability, manufacturability, and commercialization readiness.

Figure 3. High-rate lithium-ion drone batteries in pouch cell configuration.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- Designing optimal three-dimensional electrode architectures using data- and ML-assisted approaches, linking electrode geometry to electrochemical performance targets (energy density, C-rate capability, and cycle life). - Conducting simulation-driven optimization of laser processing parameters, including pattern geometry, depth, spacing, and throughput, to enable precise and repeatable electrode structuring. - Implementing laser patterning of silicon-based anodes provided by the Korean SME partner, using in-house laser processing facilities compatible with roll-to-roll (R2R) manufacturing concepts. - Developing and validating integrated R2R laser processing workflows, supported by inline sensing and adaptive process control to ensure uniformity and reproducibility. - Performing rapid prototyping and feedback-driven iteration of electrode designs, and delivering structured electrodes and technical performance data to the Korean partner for cell-level validation.
Expected R&R of Korean SME Partner	- Developing and optimizing Si/C or SiOx-based anode slurries with silicon contents of 10–30 wt%, including material selection, mixing protocols, and rheological control for thick-electrode coating. - Manufacturing silicon-based anode electrodes with target thicknesses of 40–100 μm, including coating, drying, calendaring, and thermal processing, using industrially relevant equipment. - Performing electrode slitting and notching to meet the dimensional requirements of drone batteries (25 × 35 mm^2) and supplying these electrodes to the University of Toronto for laser structuring. - Receiving laser-patterned electrodes and assembling full-cell pouch-type drone battery samples using multiple commercial cathodes (LCO, NCA, NCM). - Conducting electrochemical performance evaluation, including rate capability (up to 5C), cycle life (≥100 cycles), and dimensional stability, and providing systematic feedback to guide iterative electrode and process optimization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project aims to develop and industrially validate laser-structured silicon-based anodes that overcome the intrinsic energy–power trade-off of conventional lithium-ion batteries, with a primary focus on high-performance drone applications. **(1) Technological Outcomes** Developed IP during the project will be delivered from U of T to Remplir. This would provide a first-mover advantage by securing technological outcomes positioned to lead the rapidly expanding drone battery market. Furthermore, the engineering-based cutting-edge academic papers can be achievable during the project **(2) Commercialization Potential** The proposed technology is directly aligned with commercial lithium-ion drone battery manufacturing, as it retains conventional slurry coating, calendaring, and pouch cell assembly processes, with laser structuring implemented as an inline step. Successful outcomes will enable Remplir to commercialize high-energy, high-power drone batteries that exceed the performance limits of graphite and low-silicon blended anodes. The results are expected to strengthen Remplir's competitiveness in drone and UAV markets requiring simultaneously long flight time and high discharge capability, while providing a scalable pathway for future applications in robotics, defense UAVs, and other high-power portable energy systems. **(3) Broader Industrial and Economic Impact** At the industrial level, this project bridges the gap between academic structured-electrode concepts and real-world battery manufacturing, establishing laser-based electrode architecture as a practical process technology. Economically, the project contributes to value creation in advanced battery manufacturing, reinforces Korea–Canada collaboration in energy storage innovation, and promotes a process-driven approach to performance enhancement that reduces reliance on costly or supply-constrained materials. This project is expected to accelerate the industrial adoption of silicon-based anodes and provide a scalable solution to achieving high energy density and high-power density within a single lithium-ion battery system. In addition, the successful development of China-independent drone batteries based on Korea-Canada technologies from this project can result in diversified battery supply chain with high security and economic impact.

6. References

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